

Complements of Operational Research

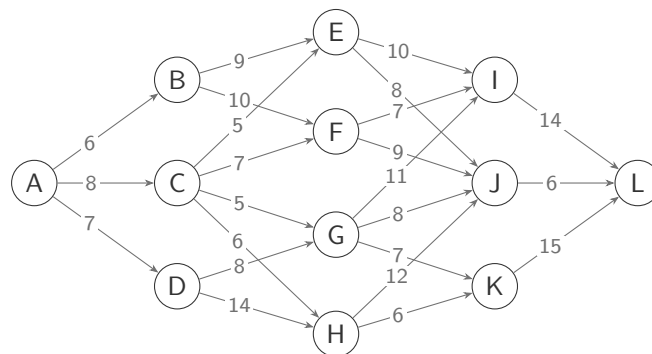
Exercise sheet 5 — Dynamic programming

The exercises marked **Homework** are to be solved at home and sent by email to the lecturer, by the deadline given in class; the solution is shared afterwards. The others come from the DP exercise sheet on Moodle or from previous tests and exams. In all of them: state the stages, states, decisions and the recurrence function.

Exercise 1 — Cheapest route from A to L

Homework

In the following graph the nodes represent towns and the edges show the travel costs (in m.u.). Using dynamic programming, find the cheapest route to travel from A to L.



Exercise 2 — Workforce planning

Homework

A building contractor estimates that, over the next 6 weeks, they need the number of workers given in the table. Surplus labour costs 300 m.u. per worker per week, and a new hiring, in any week, has a fixed cost of 500 m.u. plus 200 m.u. per worker hired. What is the optimal number of workers in each week? Solve with dynamic programming.

Week	1	2	3	4	5	6
Workers required	3	5	4	6	8	3

Exercise 3 — Knapsack problems

Homework (exam, 11/07/2019)

Solve the following knapsack problems (maximise the value placed in the knapsack) with dynamic programming, considering a maximum capacity of 10 kg.

a) Given that only one product of each type can be taken:

Product	Weight	Value
A	2	5
B	4	7
C	7	12

b) Given that more than one product of each type can be taken:

Product	Weight	Value
A	6	7
B	2	5
C	5	12

Exercise 4 — Energy-efficiency measures in a hospital

Homework (new)

A hospital has 12 (thousands of €) available for energy-efficiency measures. Each measure is carried out once or not at all. The costs and the expected annual savings are:

Measure	Cost (thousands €)	Saving (hundreds €/year)
M1 duct insulation	3	15
M2 heat pump	5	19
M3 solar panels	6	9
M4 occupancy sensors	2	8
M5 variable-speed drives	2	9
M6 LED lighting	2	19

- a) Which measures should be carried out to maximise the annual saving? Solve with dynamic programming.
- b) Which measures would the criterion “highest saving per euro invested” choose? Compare.

Exercise 5 — Knapsack with one unit of each item (I)

Moodle DP sheet (Ex. 13)

A knapsack holds 10 kg. The aim is to maximise the profit, with only one unit of each item available.

Item	Weight	Value
A	2	3
B	3	6
C	6	9

Exercise 6 — Knapsack with one unit of each item (II)

Moodle DP sheet (Ex. 14)

A knapsack holds 8 kg. The aim is to maximise the profit, with only one unit of each item available.

Item	Weight	Value
A	1	2
B	2	5
C	4	6
D	5	10
E	7	13
F	8	16

Exercise 7 — 23 kg knapsack

Moodle DP sheet (Ex. 15)

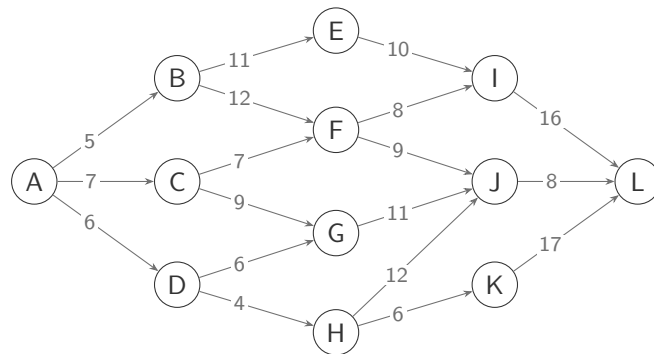
Which products should we put in the knapsack so as to maximise the profit, given that the knapsack only holds 23 kg and that only one product of each type can be taken?

Product	Weight	Value
A	1	1
B	2	6
C	5	18
D	6	22
E	7	28
F	9	40
G	11	60

Exercise 8 — Cheapest routes from A to L (I)

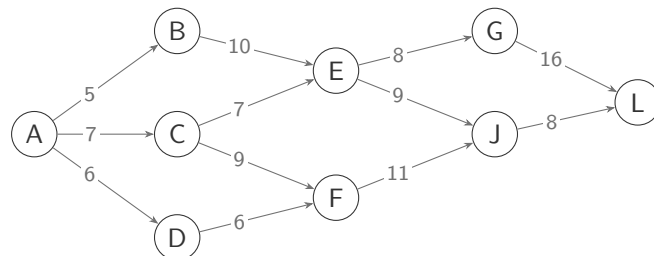
exams, 10/07/2018 and 02/07/2024

In the following graph the nodes represent towns and the edges show the travel costs (in m.u.). Using dynamic programming, find the cheapest routes to travel from A to L.

**Exercise 9 — Cheapest route from A to L (II)**

2nd test and exam, 27/06/2019

In the following graph the nodes represent towns and the edges show the travel costs (in m.u.). Using dynamic programming, find the cheapest route to travel from A to L.

**Exercise 10 — Knapsack (several units of each product)**

2nd test, 01/06/2018; exam, 02/07/2024

Using dynamic programming, solve the knapsack problem so as to maximise the value placed in the knapsack, given that its maximum capacity is 10 kg and that more than one product of each type can be taken.

Product	Weight (kg)	Value (€)
A	4	11
B	2	5
C	5	13

Exercise 11 — Workforce in a construction company

2nd test, 11/06/2024

A construction company needs to hire, over the next 5 weeks, the number of staff given in the table. Surplus labour costs 200 m.u. per worker per week, and a new hiring in any week has a fixed cost of 400 m.u. plus 150 m.u. per worker. What is the optimal number of employees in each week, with the aim of minimising the expenditure? Solve using dynamic programming.

Week	1	2	3	4	5
Workers required	6	8	9	3	5